

ANANDA WUTWANNA

• IoT & Embedded Systems Engineer

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SUMMARY

Engineer who analyzes, designs, and builds systems around what they actually need to do — across both software and the hardware it controls. On the hardware side, writing the programs that drive devices such as motors, valves, and sensors, and handling the communication between them; on the software side, building the frontend and backend, the APIs, and the platforms that command and monitor whole systems in real time. Able to take a system from gathering requirements and designing the architecture, through writing the code, to installing and troubleshooting it on-site.

EDUCATION

Bachelor of Engineering in Computer Engineering — Naresuan University

Aug 2016 — Aug 2020

WORK EXPERIENCE

Civic Agrotech — IoT Imagineer

Oct 2025 — Present

Design and develop the automation for **PFAL — a vertical farming plant factory** that moves and grows crop trays without manual handling: working out what systems the plant needs, building them, and tying them together under one control platform.

- **Tray-transport automation:** moves crop trays in and out of the growing racks by coordinating the RGV (a tray vehicle running on each growing layer's rail) and the lift that carries trays between floors — with real-time monitoring that tracks where each machine is at every step.
- **Central control platform** — the brain of the system: commands and coordinates every subsystem, runs tray put / get / move tasks from the crop plan, and monitors devices and the growing environment in real time.
- **Nutrient-water systems:** controllers for the refill valves (automatic open/close with live status) and the nutrient mixing / dosing system, dispensing to a target concentration with precise pump control.
- **Grow lights:** per-tray control of white/red LED lights and fans with automatic scheduling.
- Authored technical documentation and grow-planning analysis for the facility.

Advisory — Software Analyst

2025 — Oct 2025

- Collaborated with PO/BA and UX/UI designers to evaluate technical feasibility of system requirements.
- Prepared sections of the SRS (swimlane diagrams, API documentation, ER diagrams); defined and analyzed system issues; designed and tested standards and solutions.
- Planned and estimated resources across the software development lifecycle; verified deliverables through system testing; provided user training and operational support.

ADC Microsystems

2020 — 2025

Software Analyst

2022 — 2025

- Analyzed and designed systems from client requirements into practical solutions.
- Researched technical approaches to improve systems and meet user demands.
- Designed appropriate system architectures and coordinated with the development team to ensure system-design compliance.

Project Coordinator

2021 — 2022

- Coordinated a small development team across multiple concurrent projects.
- Analyzed and developed projects from client requirements into practical systems.
- Coordinated between development teams and clients to ensure on-time project delivery.

IoT Developer

2020 — 2021

- Developed and deployed IoT solutions from prototype to production.
- Oversaw development through deployment and installation phases.
- Conducted testing and troubleshooting in real operating environments.

TECHNICAL SKILLS

PROTOCOLS	MQTT	Modbus	RS-485	I2C	UART/Serial	TCP/UDP	WebSocket	REST	ADC
FULL-STACK & DB	React	TypeScript	Python (FastAPI)	Node.js	REST API design	PostgreSQL	MongoDB	MySQL	
MCU	ESP32	Arduino	Sensor integration	OTA updates					
EMBEDDED / RTOS	FreeRTOS	ESP-IDF	Linux (basic)	NVS					
FIRMWARE (C/C++)	Stepper motor control	PWM pump control	Device communication						
TOOLING	PlatformIO	Arduino IDE	Node-RED	VS Code	Docker	Git/GitLab CI			

SELECTED PROJECTS

Vertical Farming Plant Factory (PFAL) AutomationCivic Agrotech

Designed and built the full automation for a vertical farming plant factory — tray transport with RGVs and a lift, a central control platform, nutrient-water systems, and grow lights — tied together under one control platform with real-time monitoring.

ESP32	ESP-IDF	FreeRTOS	C/C++	Stepper motor control	Modbus/RS-485	I2C	PWM	OTA	MQTT	React	FastAPI
MongoDB											

Water Bill Management SystemADC Microsystems

Analyzed requirements and designed a water-utility management platform for local government agencies — POS terminals for field agents, a web administration portal, and a Line OA consumer interface, with RFID/NFC identification, OCR meter reading, and real-time payments. Designed the database schema and workflows, wrote the functional specifications for all three modules, and coordinated the development team.

React Native	React JS	Line OA	PostgreSQL	RFID/NFC	OCR
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Air Quality Monitoring SystemADC Microsystems

Designed a real-time air-quality platform for local agencies and universities — an IoT sensor network (PM2.5, PM10, temperature, humidity) reporting over MQTT into a web dashboard, with AQI calculation, Line OA hazard alerts, and exportable reports.

React JS	Node.js	MySQL	MQTT	IoT sensors
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Beacon-based Public Announcement SystemADC Microsystems

Designed and developed a location-aware announcement system using BLE beacons for indoor positioning, integrated with the Line Messaging API to automatically notify users near a beacon, with an engagement-analytics dashboard.

React JS	Node.js	PostgreSQL	BLE	Line API
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Smart Watch ServicesADC Microsystems

Led development of an IoT healthcare backend ingesting real-time data from multiple smartwatch models over UDP/MQTT — live health metrics (heart rate, blood pressure, SpO2, temperature), a data-processing pipeline, and an SOS emergency system with GPS/Wi-Fi location tracking.

Node.js	WebSocket	MQTT	UDP	Node-RED	PostgreSQL
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Energy Management SystemADC Microsystems

Designed a system that collects and analyzes energy-consumption data in real time from energy meters and smart devices (smart plugs / switches) over MQTT, with a dashboard for viewing and analyzing usage down to the individual device.

React JS	Node.js	PostgreSQL	MQTT	Energy meters	Smart plug/switch
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